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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS

International General Certificate of Secondary Education

MARK SCHEME for the October/November 2009 question paper for the guidance of teachers

0625 PHYSICS

0625/06 Paper 6 (Alternative to Practical), maximum raw mark 40

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1 (a) (i) *d* 0.5 cm or 5mm

(ii) *x* 10.0

(b) (i)–(iii)
table: *T* 1.0, 0.95, 0.895 (0.90, 0.9), 0.84, 0.775 (0.78) [1]
T^2 1.00, 0.903, 0.801, 0.706, 0.601 (if *T* correct) [1]

(c) graph:
axes labelled [1]
scales suitable, plots occupying at least half grid [1]
plots all correct to ½ square [1]
well judged line [1]
thin line, 5 neat plots [1]

(d) statement NO and not through origin/
inverse/negative gradient/
x increases, T^2 decreases/ wtte [1]

[Total: 10]

2 (a) 91 (^{o}C) [1]

(b) *t* in s, both θ in ^{o}C [1]

(c) statement B and justified by reference to readings [1]

(d) any two from:
same starting temperature/temperature of hot water
constant room temperature/keep away from draughts/out of direct sunlight
same time intervals [2]

[Total: 5]

3 (a)–(c)
table:
V, A, Ω [1]
V 1.8 [1]
I 0.25 [1]
R values 7.20, 3.46(3.5) [1]
consistent significant figures for *R* (2 or more) [1]

(d) *y* 0.48, 0.49, 0.5 (ecf) [1]
2/3 significant figures and no unit [1]

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(e) **(i)** correct symbols and circuit (ignore power source symbol)

(ii) voltmeter position correct

(iii) control current/voltage/resistance/speed of motor [1]

[Total: 10]

4 **(a)** *f* 14.95 ± 0.05 (cm) [1]
unit to match number [1]

(b) more than one value shown [1]
d 6.5 ± 0.1 [1]

(c) *t* 0.85 ± 0.05 (cm) [1]
d and *t* both with correct unit [1]

(d) diagram showing blocks correctly placed [1]
rule shown correctly touching both blocks [1]

(e) *f* 10.9 – 13.1 (cm) (or 109 – 131 (mm)) [1]
no, too far out to be explained by experimental inaccuracy (wtte) [1]

[Total: 10]

5 **(a)** lens between object and screen (not mirror) [1]
lens at least 2 cm from object and screen [1]
metre rule on bench or clamped [1]

(b) any two from:
use of darkened room/brighter object
slowly moving lens back and forth to obtain good image
avoid parallax, action given
lining up object and lens
object and lens at same height from bench/object on principal axis
repeats
screen/lens perpendicular to bench
mark centre of lens position on block [2]

[Total: 5]